STANDARD COLLEGE NTUNGAMO

END OF TERM ONE 2026

S5 PHYSICS PAPER 1

TIME : 1 HOUR 40 MINUTES

INSTRUCTIONS: Attempt all questions

SECTION A : PARTICLES

ITEM 1

At a research laboratory studying radioactive minerals in western Uganda, a team of physicists discovers a newly extracted ore suspected to emit different types of nuclear radiation. To analyze the emissions, they set up an ionization chamber connected to a sensitive ammeter.

During the experiment, the following observations are made:

When the radiation passes through a sheet of paper, part of it is completely absorbed.

Another portion passes through paper but is stopped by a thin aluminum sheet.

A third component penetrates both paper and aluminum but is significantly reduced by a thick lead block.

The ionization chamber shows varying current readings depending on the type of radiation entering it. It is also observed that some radiations produce denser ionization tracks inside the chamber than others.

TASK

(a) Identify the three types of radiation emitted from the radioactive ore.

(b) For each type of radiation identified, state two characteristics.

(c) Explain how the ionization chamber detects radiation.

(d) The half life of Thorium-234 is 5.45 years. Determine the activity of 2g of Thorium. (Avogadro constant is 6.02×10^{23} per mole)

SECTION B : FORCE AND MOTION

ITEM 2

At a community innovation project in a rural town, a group of senior six students is designing a small wind turbine system to generate electricity for nearby homes.

During testing, they observe that the power output, **P** of the turbine depends on the blade length **l**, wind speed **v**, and air density **ρ**. They propose that these quantities are related through a dimensionless constant.

As the turbine operates, the rotating blades experience a resistive drag force, **F** due to air. This force is modeled by the expression:

$F = \frac{1}{2} C \rho A v^2$ where **A** is the effective area of the blades and **C** is not known.

During installation, a uniform ladder of mass **20 kg** and length **1 m** is used to support part of the structure. The ladder rests on rough horizontal ground and leans against a smooth vertical wall. A **10 kg** component of the turbine is temporarily attached to the ladder at a point $\frac{3}{4}$ of its length from the base.

The coefficient of friction between the ladder and the ground is **0.5**.

HINT

Acceleration due to gravity, **g=9.81ms**$^{-2}$ and that the ladder remains in equilibrium under the action of forces and moments.

TASK

(a) Derive the relationship between P and other relevant quantities.

(b) State the nature of C.

(c) State the principle governing the rotational stability of the ladder.

(d) Determine the angle the ladder makes with the ground.

SECTION C: ENERGY

ITEM 3

A team of technicians from an optics laboratory was testing a newly manufactured plane mirror. They directed a narrow beam of light from a fixed lamp through a transparent screen onto a mirror, the light was reflected. When the mirror was rotated slightly through a certain angle, the reflected light shifted through a larger angle than the mirror itself. One of the technicians informed the others that the relationship between the reflected ray and the mirror's rotation is an important concept with many real-life applications in solving many problems, though they did not know how the relationship can be obtained since they are interested in designing a device that is used in the field of astronomy to measure the angle of elevation of celestial bodies like sun and stars.

Task:

As a student of physics;

(a) Identify the laws that governing reflection of light in the above scenario.

(b) Explain the type of reflection taking place with reference to the above scenario.

(c) Help the technicians to understand the whole process of obtaining the relationship in reference to the above scenario.

(d) If the plane mirror was rotated through an angle of 20°, determine the angle through which the reflected ray will be rotated.

(e) Identify and explain the working of the device that technicians are interested in designing.

END